

(like triboelectric generators) or useful chemical reactions. These systems can be commonly employed in wearables, or other such applications.

The third class is of systems that rely on mechanical energy. This includes vibrational energy harvesters, electrodynamic energy harvesters, and others. Vibrational energy harvesters can be integrated in various wireless systems and electrodynamic energy harvesters are mostly employed in portable electronic devices like radio, flashlights, phone chargers, etc.

Table 1 gives an idea of which harvesting technique is best advised for a certain application.

Implemented battery-less solutions

Now that the popular energy harvesting techniques are familiar, it is important to identify the reason for which these can be utilised for diverse applications. To get an idea about that, we learn from some of the most innovative products and technologies that employ energy harvesting techniques, eliminating the use of batteries.

Among all the techniques discussed, RF energy harvesting is most widely used. A fabless IoT company, Wiliot, in 2021 released its Bluetooth tag for easy tracking of shipments such as food and medicine. The IoT device can harvest energy from ambient RF signals and convert it into electricity for powering circuits. This tag allows

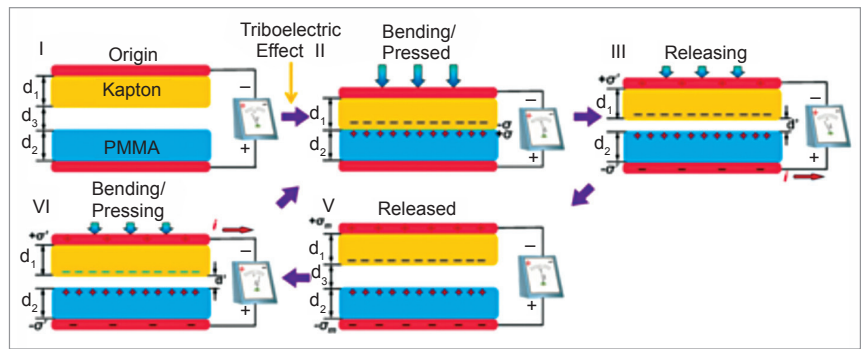


Fig. 4: Working of triboelectric nanogenerators

customers to see all aspects of the item's journey, for example, inclusive of the freshness of food.

The RF harvester, in this case, supplies an ARM microcontroller with a 3-core CPU, RAM, and ROM. The device uses three looped antennas for harvesting more energy and also takes advantage of the nanowatt power consumption of the ARM CPU during sleep mode. However, for power hungry devices, either the energy harvested should be increased or a reduction should be done in the circuit's power consumption. If both these factors are looked into, the system can be highly reliable.

Coming to vibrational energy harvesters, researchers at MIT have developed a method for underwater communication that utilises piezo elements without any power source. The system utilises a master/slave model where the master has a power source

that transmits acoustic signals, and the slave device detects the wave and uses its piezo transducers to generate power from the acoustic signal.

Thermoelectric and triboelectric harvesters are currently being employed in various wearable applications. Thermoelectric converters can be seen in smart watches and fitness trackers. Researchers are fabricating triboelectric generators inside fabrics, shoes, and other wearables for harvesting energy.

While choosing energy harvesters, the energy consumption of the application should be well predicted. There are solutions that help you build your platform or system from energy harvesters. For example, On Semiconductor's RSL10 is a multi-sensor platform based on ultra-low-power solar cells. It integrates a low-profile 47μF storage capacitor onboard for energy storage. Its power consumption is just

Best Harvesting Techniques for Various Applications

Table 1

Energy Source	Harvesting Device	Achievable Power Density	Maximum Achievable Power Recorded (With Optimised Circuits)	Feasible for
Solar (Indoor)	Solar Cells	0.006mW/cm ² (standard office desk) 0.057mW/cm ² (<60W desk lamp)	~100mW	Calculators, Wireless Sensor Networks (WSNs), Outdoor and Indoor Monitoring Systems like Weather Stations
Vibrations	Electrostatic Electromagnetic Piezoelectric	10-100μW/cm ²	430μW 493μW at 13Hz 963.9μW at 17Hz with 3g acceleration	IoT Networks, Zigbee, GPS Trackers, Hearing Aids, Pacemakers, Analogue Clocks, Timer ICs
Acoustic Noise	Piezo-transducers	3E-6mW/cm ² at 75dB 9.6E-4mW/cm ² at 100dB	~10mW	Communication Systems, Wearables
RF Signals	RF Energy Harvesters	10-100μW	>1mW	Wearables, IoT Sensor Networks, GPS Trackers, Communication Systems, Bluetooth and RF Tags, Backscatter Communication Systems
Static Electricity	Triboelectric Nanogenerators	10-30μW/cm ²	Around 100-200μW	Wearables
Thermal Energy	Thermoelectric Generators	In the range of mW	~1W	Wearables
Mechanical work	Electrodynamic Systems	In the range of mW	In the order of few watts	Small-scale Electronics like Radio, Flashlights, etc